Infosys - Coding 8

HackerRank

You are given an array A of size N and Q queries. Each query consists of two numbers (pos, x), where:

pos: An index in A (0-based).

x: A bit position (0 to 19).

For each query, we need to count increasing subsequences that end at A[pos], such that the XOR of the subsequence has the x-th bit set to 1.

Find the sum of results of all queries modulo 10^9 + 7.

Input Format

The first line contains an integer, N, denoting the number of elements in A.

The next line contains an integer, Q, denoting the number of rows in Queries.

Each line i of the N subsequent lines (where 0 ≤ i < N) contains an integer describing A[i].

The next line contains an integer, Cols, denoting the number of columns in Queries.

Each line i of the Q subsequent lines (where 0 ≤ i < Q) contains Cols space separated integers each describing the row Queries[i].

Constraints

1 <= N <= 500000

1 <= Q <= 10^5

1 <= A[i] <= 2^20 - 1

1 <= Cols <= 2

0 <= Queries[i][j] <= N-1

Output Format

The output contains an integer value

Sample Input 0

```
2
1
1
2
2
1 1
```

Sample Output 0

```
2
```

Explanation 0

N=2, Q=1, A=[1,2] Query: (1,1) → Find increasing subsequences ending at A[1] (i.e., 2) where the XOR has the 1st bit set. Possible subsequences: [2], XOR = 2 → Binary: 10 (1st bit is 0, not counted). [1,2], XOR = 1 ⊕ 2 = 3 → Binary: 11 (1st bit is 1, valid). Result for this query = 1. Final Output: 2.

Sample Input 1

```
2
1
1
2
2
0 0
```

Sample Output 1

```
1
```

Explanation 1

N = 2, Q = 1, A = [1, 2] Query: (0, 0) → Find increasing subsequences ending at A[0] (1) where the 0-th bit is set. Checking Possible Subsequences: [1] → XOR = 1 → Binary: 1 (0-th bit is 1) This sequence is valid.